

الإجابات على "كراس الإجابة"

I- Choose the correct answer: (5 pts)

1- Choose the algorithm which deletes the last cell of a non-empty linear list named LIST.

a- PROCEDURE: DELETE (LIST)

BOX : Pointer;

BEGIN

BOX ← LIST;

LIST ← NEXT (LIST);

RELEASE (BOX);

END PROCEDURE;

b- PROCEDURE: DELETE (LIST)

L, BOX, BEF : Pointer;

BEGIN

L ← LIST;

BEF ← LIST;

WHILE NEXT(L) <> NIL

DO

BEF ← L;

L ← NEXT(L);

END DO;

BOX ← L;

RELEASE (BOX);

NEXT(BEF) ← NIL;

END PROCEDURE;

c- PROCEDURE: DELETE (LIST)

L, BOX : Pointer;

BEGIN

BOX ← LIST;

LIST ← NEXT(LIST);

RELEASE (BOX);

WHILE NEXT(L) <> NIL

DO

L ← NEXT(L);

END DO;

END PROCEDURE;

d- PROCEDURE: DELETE (LIST)

BOX : Pointer;

BEGIN

VALUE (BOX) ← VD;

NEXT (BOX) ← LIST;

LIST ← BOX;

END PROCEDURE;

2- Which expression is valid?

a- Postfixed: $CD + EF - * A 3 \uparrow / * +$

b- ECP: $(((E / C) + (A - B)) * D)$

c- Prefixed: $/ - CD * EF * \uparrow A 2$

d- Infix: $B + ES - A / C * / F$

3- What condition is essential in a recursive function?

a- An affectation

b- A call to another function

c- A base case (condition) to stop

d- A loop

4- Suppose the following recursive function :

FUNCTION EXISTVAL (LIST, VAL): Boolean

IF LIST = Null

THEN EXISTVAL ← FALSE;

ELSE IF VALUE (LIST) = VAL

THEN EXISTVAL ← TRUE;

ELSE EXISTVAL ← EXISTVAL

(NEXT(LIST), VAL);

END IF;

END IF;

END FUNCTION;

Which of the following sentences is correct?

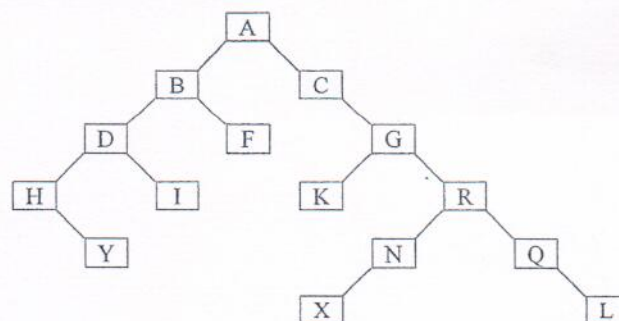
a- The function returns the rank of a value VAL in a linear list (LIST).

b- The function returns the number of times the value VAL is repeated in the list.

c- The function returns a Boolean value if VAL exists or not in the list.

d- Function is incorrect.

5- By applying the prefixed (preorder) traversal, the output of following tree is:



a- ABCDFGHIKRYNQXL

b- YHIDFBKXNLQRGCA

c- ABDHYIFCGKRNXQL

d- XLYNQHIKRD FGBCA

II- Choose and treat only 3 questions: (5pts/question)

1- Write a procedure that adds a new cell for a given value VAL at the end of a bidirectional list BLIST.

2- Convert the following expression to an expression fully (completely) parenthesized and in a tree:

$$A(C+B) < \frac{(A+B)2K}{C+B \cdot D-7} - 4^{A/B} \text{ and } (A+Y)^3 < I - B^X$$

3- Write a function that counts the number of nodes in a binary tree BTREE. The choice of the tree traversal and the method (iterative or recursive) is yours.

4- Write the algorithm named **DivisibleByN(List1 pointer, n integer)** that returns the number (counter) of cells that hold a value divisible by n (**Value MOD n = 0**). The algorithm must traverse a linear list List1.

For example : If the above linked list has the values (6, 4, 10, 9, 1) and if n is 3, then the method will return 2 (only 6 and 9 are divisible by 3).

I

Algo

1	2	3	4	5
b	b	c	c	c

II 1) Procedure Addlast (Blist: ptr, VAL: value) $\frac{1}{2}$
 var p, R: ptr $\frac{1}{4}$
 Begin $\frac{1}{4}$
 R = newnode $\frac{1}{4}$; info(R) = val; next(R) = null; $\frac{1}{4}$
 p = Blist $\frac{1}{4}$
 if (Blist <> null) then $\frac{1}{4}$
 while (next(p) <> null) $\frac{1}{4}$
 do
 p = next(p) $\frac{1}{4}$
 end do
 next(p) = R $\frac{1}{4}$
 prev(R) = p $\frac{1}{4}$
 else
 Blist = R $\frac{1}{4}$
 prev(Blist) = null $\frac{1}{4}$
 end if
 end

Method 1

1) Procedure Addlast (Blist: ptr, val: value) $\frac{1}{2}$
 var p, R, Ln: ptr $\frac{1}{4}$
 Begin $\frac{1}{4}$

 Ln = null; p = Blist; R = newnode; info(R) = val; next(R) = null $\frac{1}{4}$
 while (p <> null) $\frac{1}{4}$
 do
 Ln = p; p = next(p) $\frac{1}{4}$
 end do
 if Ln = null then $\frac{1}{4}$
 Blist = R $\frac{1}{4}$
 end if
 next(Ln) = R $\frac{1}{4}$
 prev(R) = Ln $\frac{1}{4}$
 end

Method 2

2) ECP:

$$(((A+B)^{(2 \times K)}))$$

Algo

OR for ^ $\updownarrow$

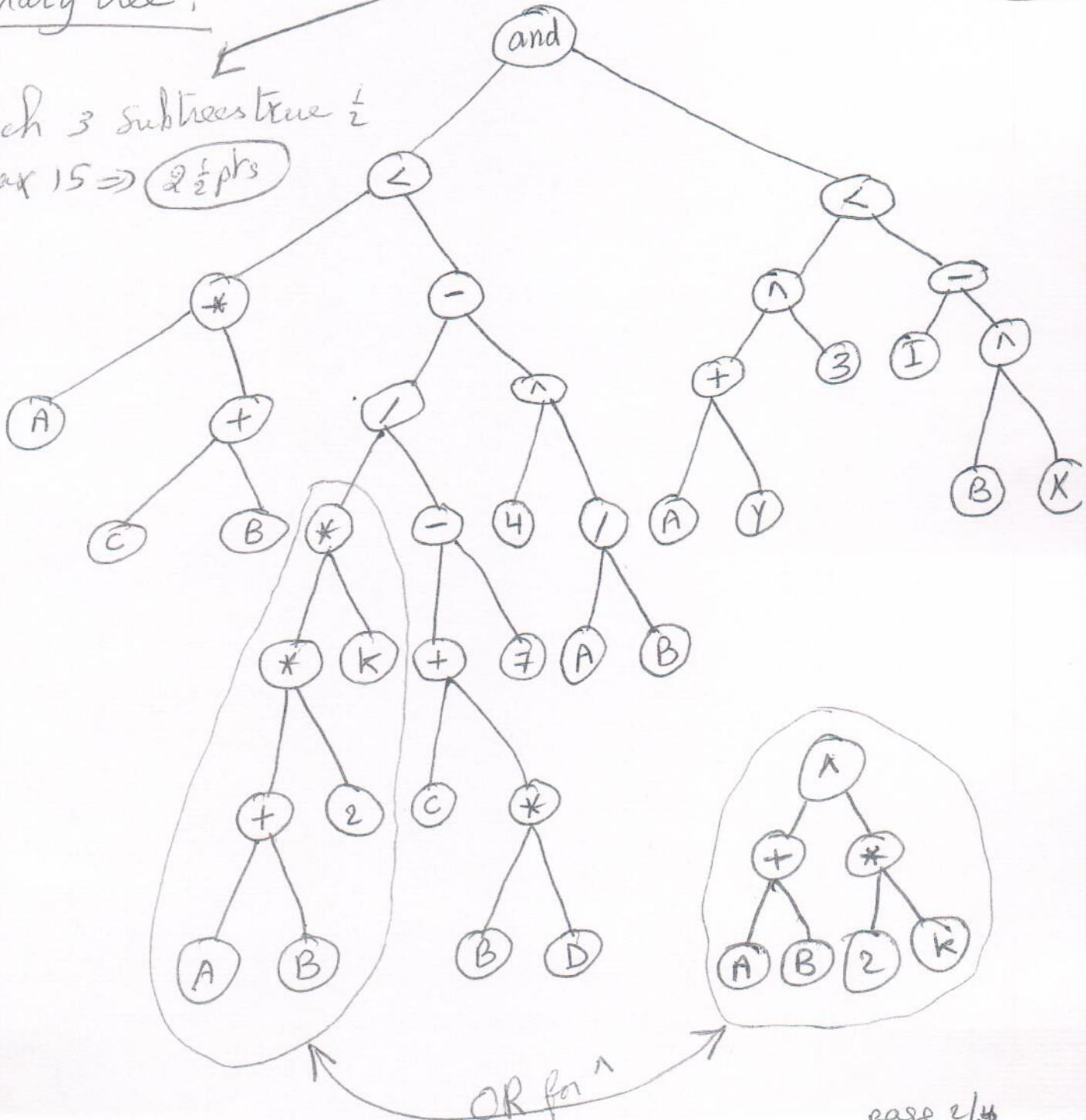
$$(((A * (C+B)) < (((((A+B) * 2) * K) / ((C + (B * D))$$

$$-7))) - (4 \wedge (A/B)))) \text{ and } (((A+Y)^3 < (I - (B \wedge X))))$$

each 3 brackets true $\frac{1}{2}$ Max 15 $\Rightarrow 2\frac{1}{2}$ pr

Binary tree:

each 3 subtrees true $\frac{1}{2}$
Max 15 $\Rightarrow 2\frac{1}{2}$ prs



Nº 3

Algo

Method 1:

~~Procedure~~ Prefix1 (Root: ptr, ctr: Integer) ^{Integer}

var s: stack, p: ptr

Begin

ctr = 0

empty(s) = true ; p = Root

① While (p <> null or not empty(s))

do while (p <> null)

do ctr = ctr + 1

push(s, p)

p = left(p)

enddo

pop(s, p)

p = Right(p)

enddo

end

return ctr

Method 3:

~~Function~~

Procedure Postfix1 (Root: ptr,

ctr: integer)

var p: ptr

s: stack, ind: boolean

integer

Begin

Empty(s) = true ; p = root

while (p <> null or not empty(s))

do while (p <> null)

do push(s, p, true)

p = left(p)

enddo

pop(s, v, ind)

if (ind = true) then
push(s, p, false)
else p = Right(p)

Integer

~~Procedure Prefix1 (p)~~

~~Begin~~

Method 2:

~~function~~

~~Procedure infix1 (Root: ptr,~~

~~ctr: integer)~~

~~var s: stack, p: ptr~~

~~in~~

Begin

empty(s) = true ; p = root

while (p <> null or not empty(s))

do while (p <> null)

do

push(s, p)

p = left(p)

enddo

pop(s, p)

ctr = ctr + 1

p = Right(p)

enddo

return ctr

ctr = ctr + 1
p = null
end if
enddo
return ctr
end

الإمضاء	الاسم	إسم اللجنة
	فاروق الحركه	رئيس
	أنطون يزبك	نائب
		المعلوماتية LT - TS

الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

تاريخ : تموز ٢٠٢٥

أسس التصحيح للمسابقة : الغوريتم وهيكلية المعطيات (٢٣١٠٥)

3 - FUNCTION : SIZE (BTREE)

IF BTREE = NIL

THEN: SIZE $\leftarrow$ 0;

ELSE: SIZE $\leftarrow$ 1 + SIZE (LEFT (BTREE)) + SIZE (RIGHT (BTREE));

END IF;

END;

4 - FUNCTION DIVISIBLEBYN (LIST1, n)

L $\leftarrow$ LIST1;

CT $\leftarrow$ 0;

WHILE L $\neq$ NIL

DO

IF VALUE (L) MOD n = 0

THEN: CT $\leftarrow$ CT + 1;

END IF;

L $\leftarrow$ NEXT (L);

END DO;

RETURN (CT);

END;

Or

FUNCTION: DIVISIBLEBYN (LIST1, n)

IF LIST1 = NIL

THEN: DIVISIBLEBYN $\leftarrow$ 0;

ELSE IF VALUE (LIST1) MOD n = 0

THEN DIVISIBLEBYN $\leftarrow$ 1 + DIVISIBLEBYN (NEXT(LIST1), n);

ELSE DIVISIBLEBYN $\leftarrow$ DIVISIBLEBYN (NEXT(LIST1), n);

END IF;

END IF;

END;

Method 2:
function PREL (Root: ptr, ctr: integer)
var s: stack; p: ptr;
Begin
ctr = 0; empty (s);
p = root;
while (p $\neq$ null or not empty (s))
do
while (p $\neq$ null)
do
ctr = ctr + 1; push (s, p)
p = left (p)
enddo
pop (s, p)
p = right (p)
enddo
return ctr
end

3/3

الإمضاء	أسماء المشاركين في التصحيح	الإمضاء	أسماء المشاركين في التصحيح
	باسم دبلار		الاسم الحوري
	سلطان		نخبة كسبي
	مينا التري		لينا
	غور		عمود